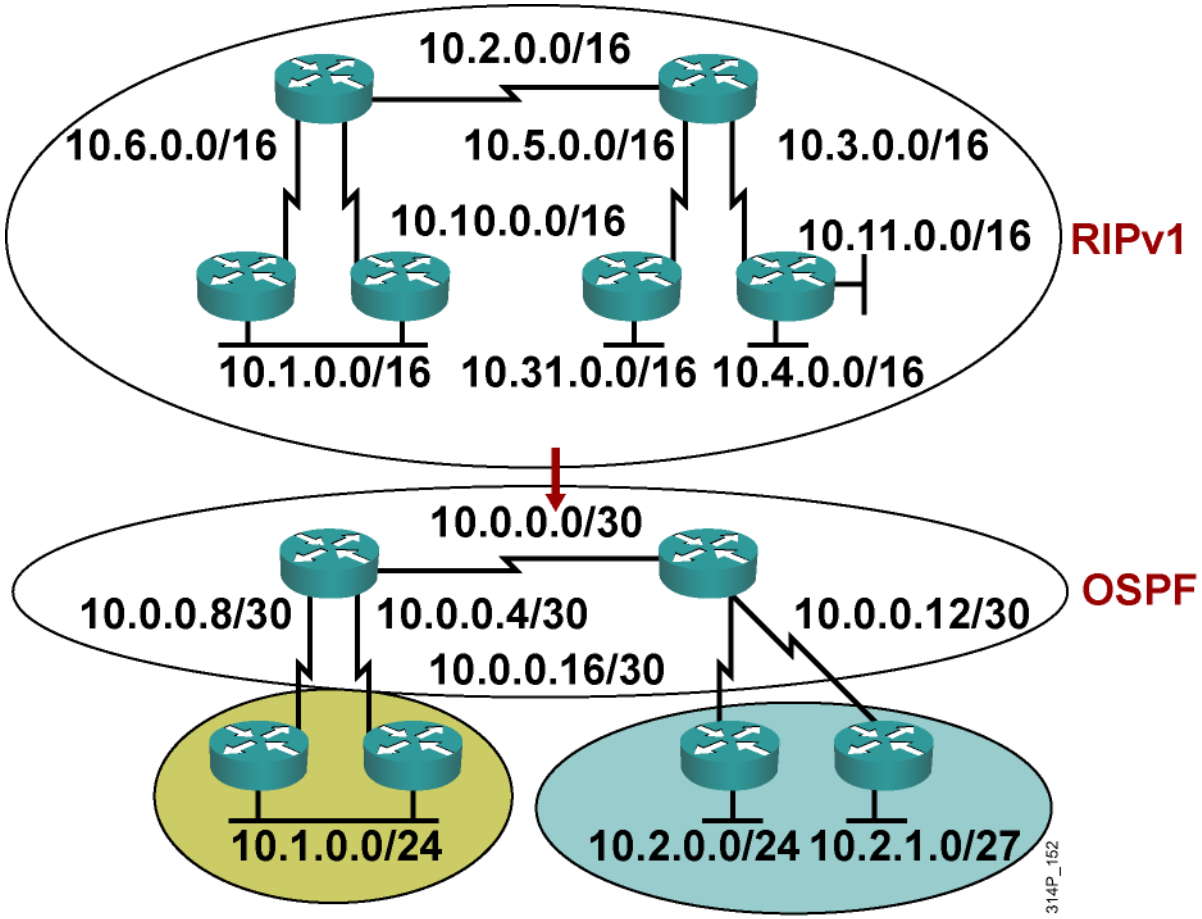


Manipulating Routing Updates

Operating a Network Using Multiple IP Routing Protocols

Using Multiple IP Routing Protocols

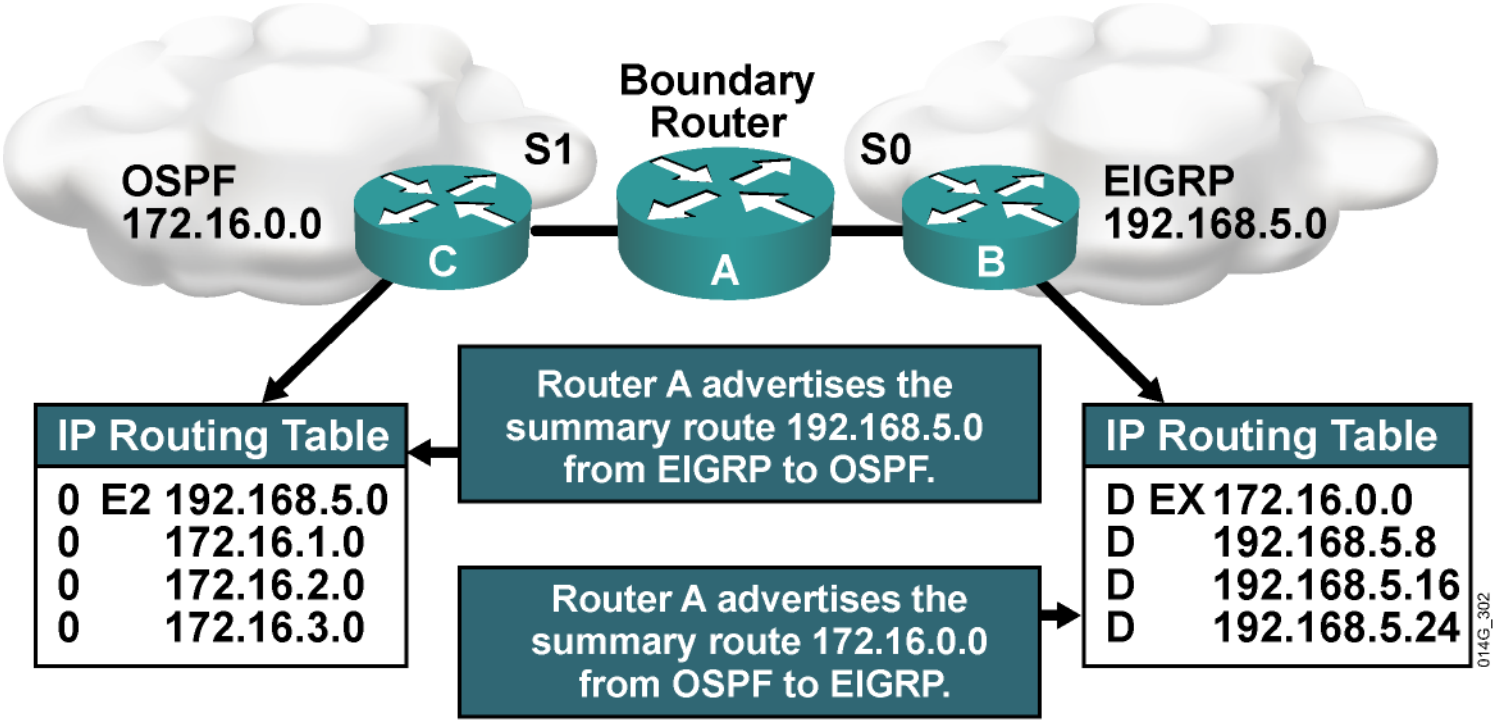


- FLSM to VLSM
- Hierarchical Addressing
- Hierarchical Areas

Using Multiple Routing Protocols

- Interim during conversion
- Application-specific protocols
  - One size does not always fit all.
- Political boundaries
  - Groups that do not work well with others
- Mismatch between devices
  - Multivendor interoperability
  - Host-based routers

Redistributing Route Information



Routes are learned from another routing protocol when a router redistributes the information between the protocols.

Using Seed Metrics

- **Use the default-metric command to establish the seed metric for the route or specify the metric when redistributing.**
- **Once a compatible metric is established, the metric will increase in increments just like any other route.**

Redistribution with Seed Metric

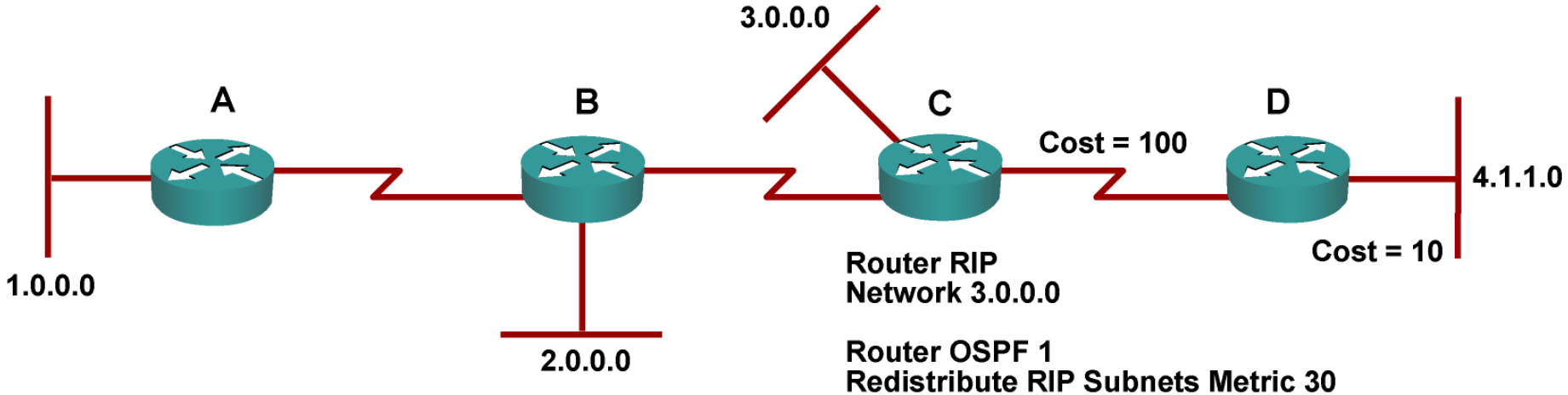


Table A

```
C 1.0.0.0
R 120/1 2.0.0.0
R 120/2 3.0.0.0
```

Table B

```
C 2.0.0.0
R 120/1 1.0.0.0
R 120/1 3.0.0.0
```

Table C

```
C 3.0.0.0
R 120/1    2.0.0.0
R 120/2    1.0.0.0
O 110/110  4.1.1.0
```

Table D

```
C  4.1.1.0/24
E2 110/130 1.0.0.0
E2 110/130 2.0.0.0
E2 110/130 3.0.0.0
```

314P_155

Summary

- **Using multiple IP routing protocols can be a result of migrating to a more advanced routing protocol, a multivendor environment, political boundaries, or device mismatch.**
- **The way that redistributed routes will appear in the routing table will vary depending on the protocols being redistributed and how they are redistributed.**
- **The seed metric is the metric associated with the redistributed route and should make the route appear worse than any internal route.**



Manipulating Routing Updates

Configuring and Verifying Route Redistribution

Redistribution Supports All Protocols

```
RtrA(config)#router rip
RtrA(config-router)#redistribute ?
  bgp         Border Gateway Protocol (BGP)
  connected   Connected
  eigrp       Enhanced Interior Gateway Routing Protocol (EIGRP)
  isis        ISO IS-IS
  iso-igrp    IGRP for OSI networks
  metric      Metric for redistributed routes
  mobile      Mobile routes
  odr         On Demand stub Routes
  ospf        Open Shortest Path First (OSPF)
  rip         Routing Information Protocol (RIP)
  route-map   Route map reference
  static      Static routes
  <cr>
```

Configuring Redistribution into RIP

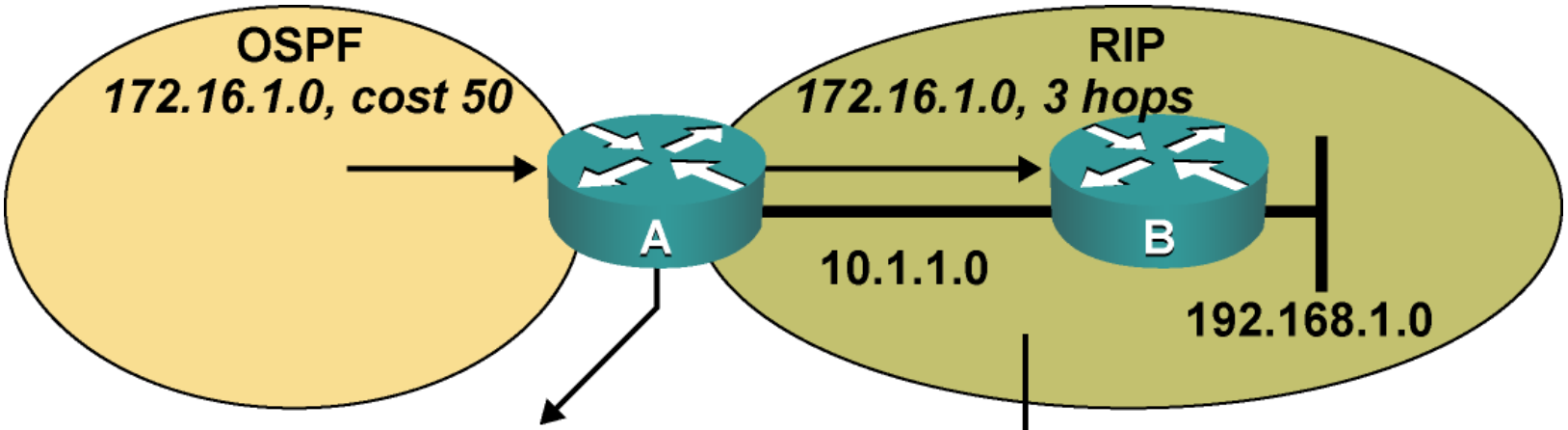
```
RtrA(config)# router rip
RtrA(config-router)# redistribute ospf ?

 <1-65535>  Process ID
RtrA(config-router)# redistribute ospf 1 ?

  match       Redistribution of OSPF routes
  metric      Metric for redistributed routes
  route-map   Route map reference
  …
  <cr>
```

Default metric is infinity.

Redistributing into RIP



router rip
redistribute ospf 1 metric 3

Configuring Redistribution into OSPF

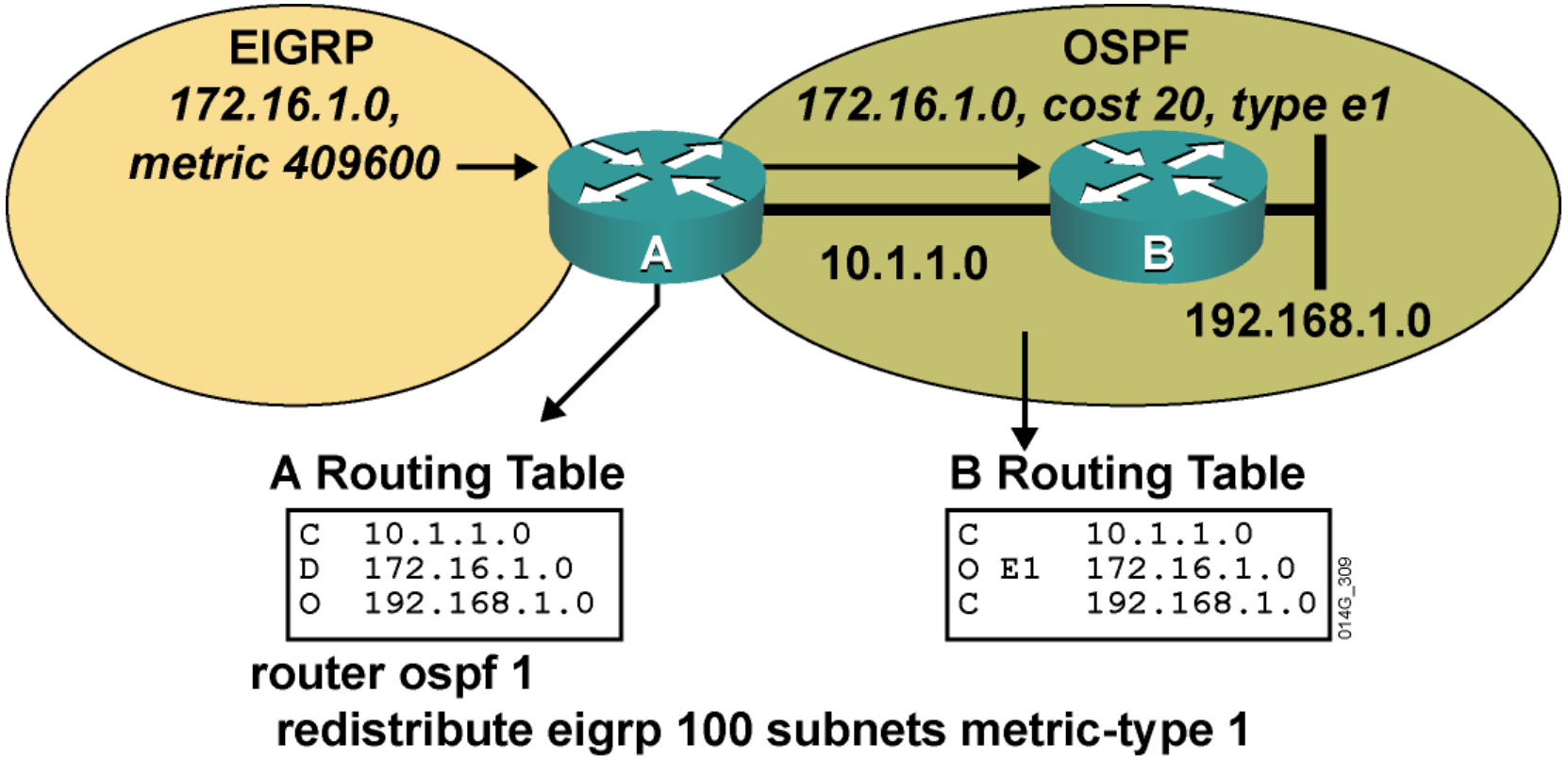
```
RtrA(config)# router ospf 1
RtrA(config-router)# redistribute eigrp ?

  <1-65535>  Autonomous system number
RtrA(config-router)# redistribute eigrp 100 ?

  metric          Metric for redistributed routes
  metric-type     OSPF/IS-IS exterior metric type for redistributed routes
  route-map       Route map reference
  subnets         Consider subnets for redistribution into OSPF
  tag             Set tag for routes redistributed into OSPF
  …
  <cr>
```

- **Default metric is 20.**
- **Default metric type is 2.**
- **Subnets do not redistribute by default.**

Redistributing into OSPF



Configuring Redistribution into EIGRP

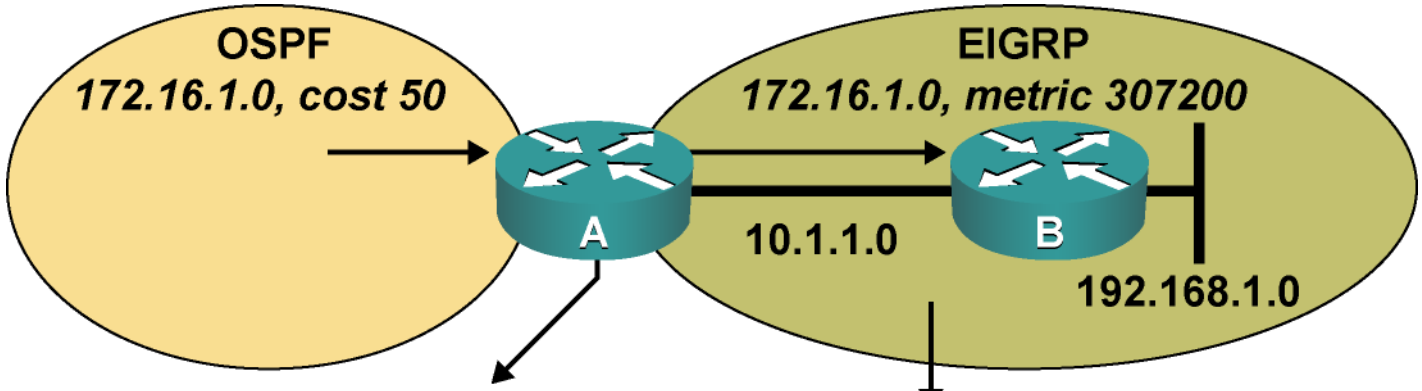
```
RtrA(config)# router eigrp 100
RtrA(config-router)# redistribute ospf ?

  <1-65535>  Process ID
RtrA(config-router)# redistribute ospf 1 ?

  match         Redistribution of OSPF routes
  metric        Metric for redistributed routes
  route-map     Route map reference
  …
<cr>
```

- **Default metric is infinity.**

Redistributing into EIGRP




```
router eigrp 100
  redistribute ospf 1 metric 10000 100 255 1 1500
```

- **Bandwidth in kilobytes = 10000**
- **Delay in tens of microseconds = 100**
- **Reliability = 255 (maximum)**
- **Load = 1 (minimum)**
- **MTU = 1500 bytes**

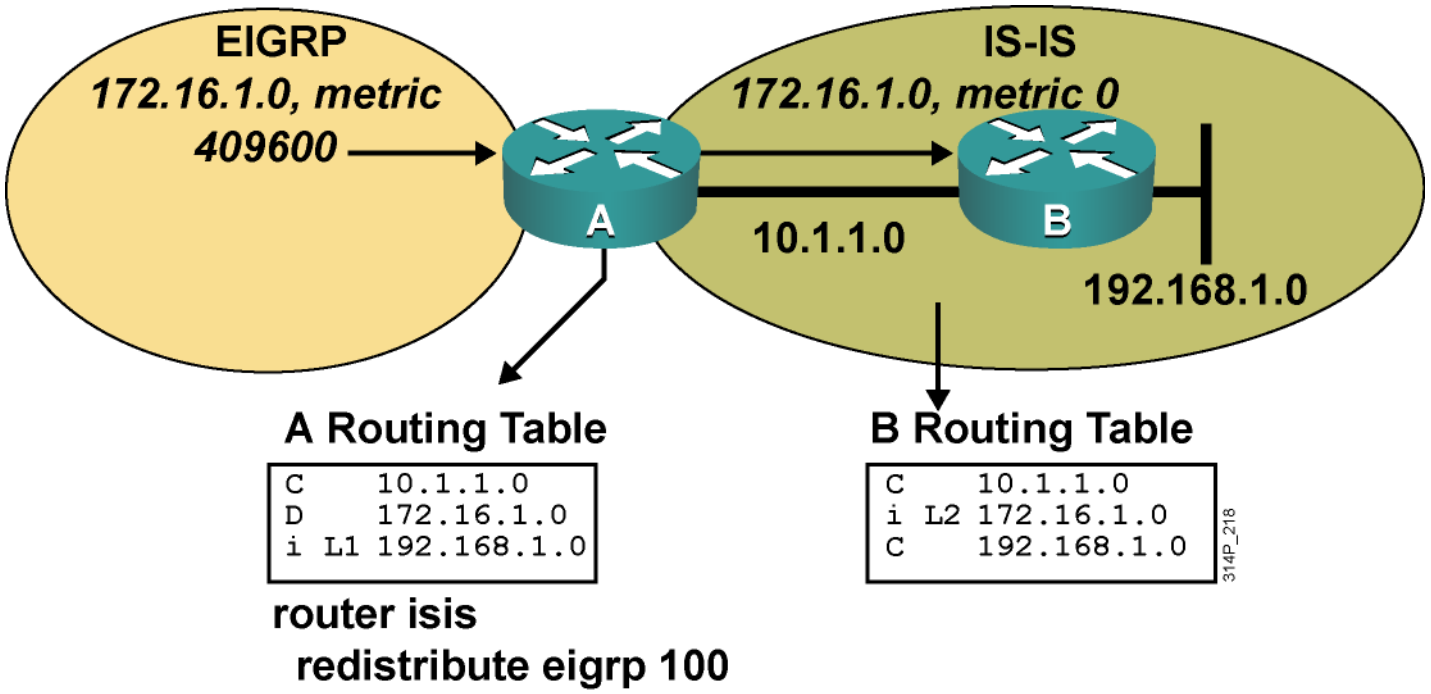
Configuring Redistribution into IS-IS

```
RtrA(config)# router isis
RtrA(config-router)# redistribute eigrp 100 ?

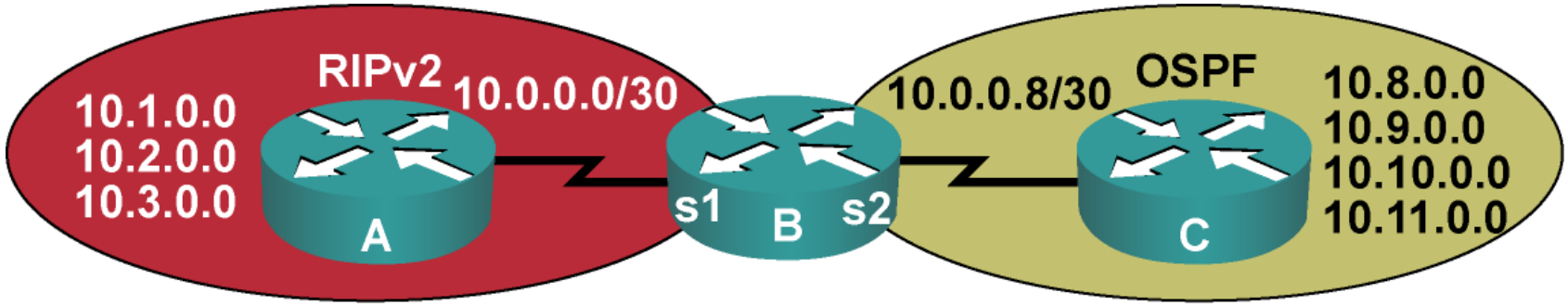
  level-1       IS-IS level-1 routes only
  level-1-2     IS-IS level-1 and level-2 routes
  level-2       IS-IS level-2 routes only
  metric        Metric for redistributed routes
  metric-type   OSPF/IS-IS exterior metric type for redistributed routes
  route-map     Route map reference
  ..
  Output Omitted
```

Routes are introduced as Level 2 with a metric of 0 by default.

Redistributing into IS-IS



Example: Before Redistribution



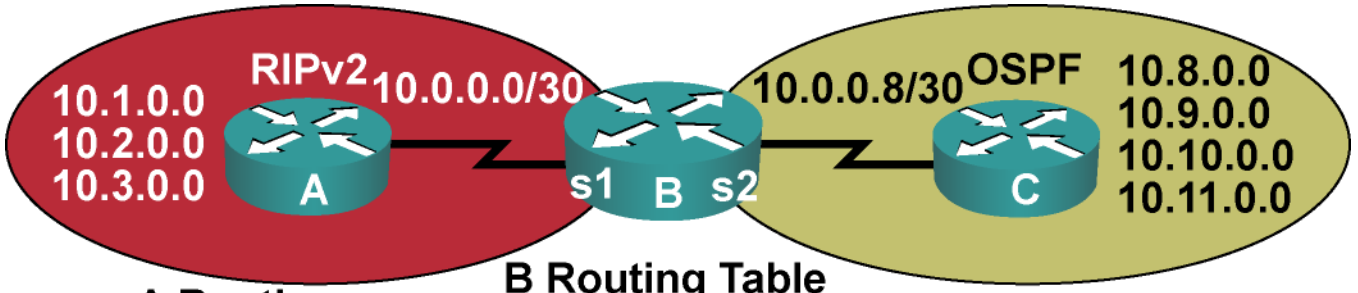
Router B Configuration

```
router ospf 1
  network 10.0.0.8 0.0.0.3 area 0

router rip
  network 10.0.0.0
  version 2
  passive-interface s2
```

014G_312

Example: Before Redistribution (Cont.)



A Routing Table

C	10.0.0.0
R	10.1.0.0
R	10.2.0.0
R	10.3.0.0

B Routing Table

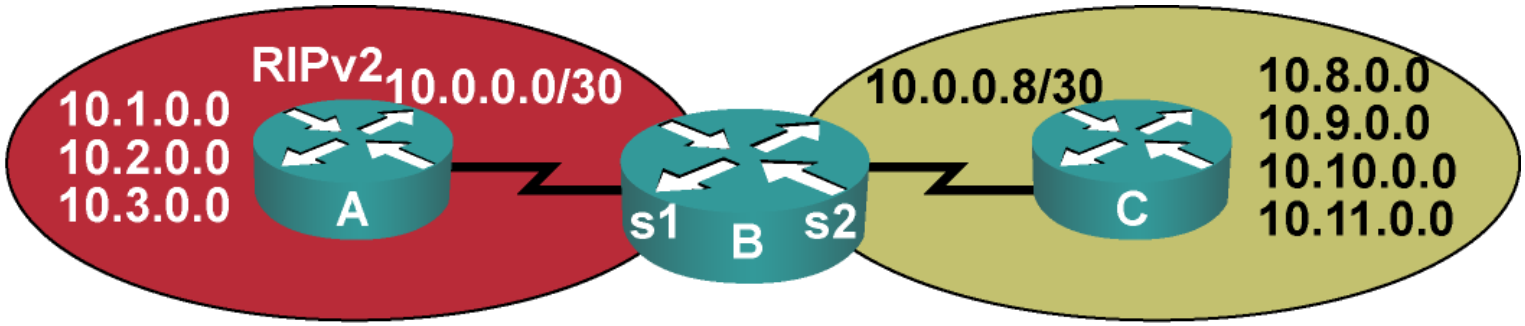
C	10.0.0.0
C	10.0.0.8
R	10.1.0.0
R	10.2.0.0
R	10.3.0.0
O	10.8.0.0
O	10.9.0.0
O	10.10.0.0
O	10.11.0.0

C Routing Table

C	10.0.0.8
O	10.8.0.0
O	10.9.0.0
O	10.10.0.0
O	10.11.0.0

314P_162

Example: Configuring Redistribution at Router B



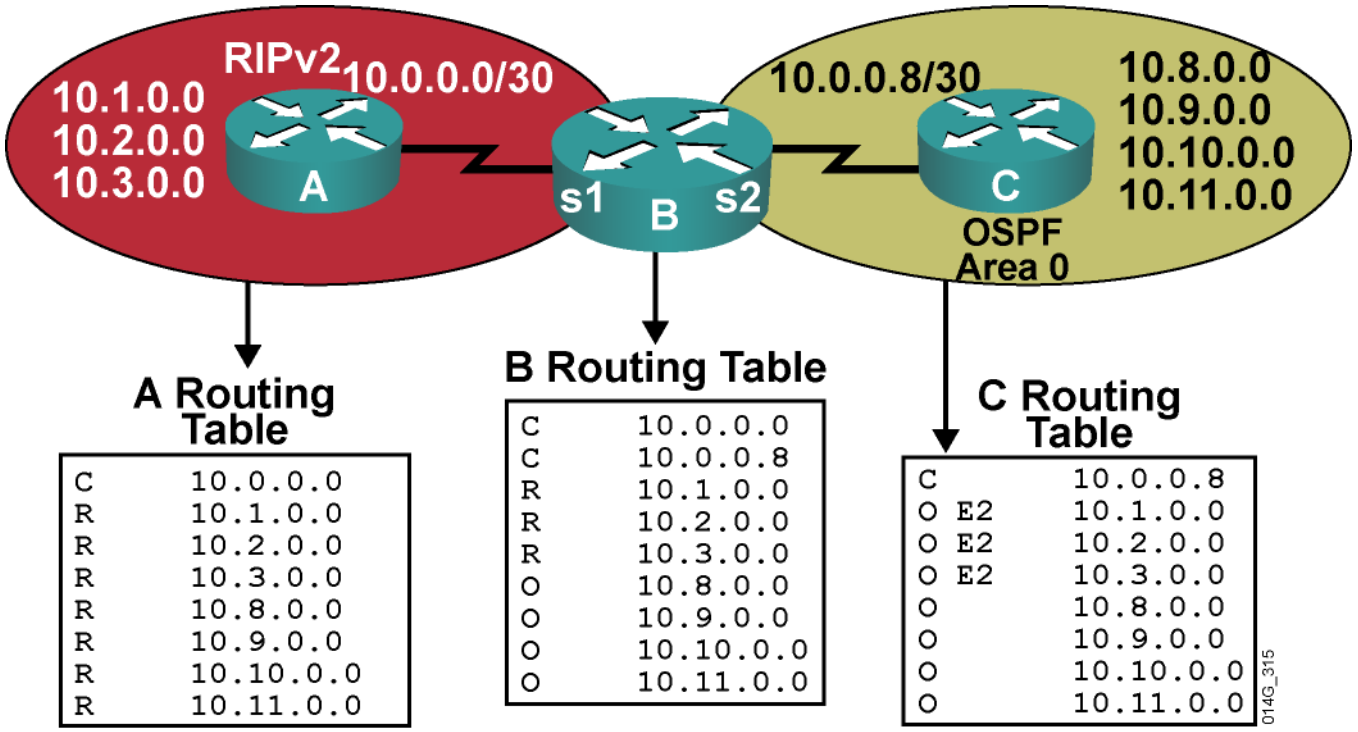
Router B Configuration

```
router ospf 1
  network 10.0.0.8 0.0.0.3 area 0
  redistribute rip subnets metric 300

router rip
  network 10.0.0.0
  version 2
  passive-interface s2
  redistribute ospf 1 metric 5
```

014G_515

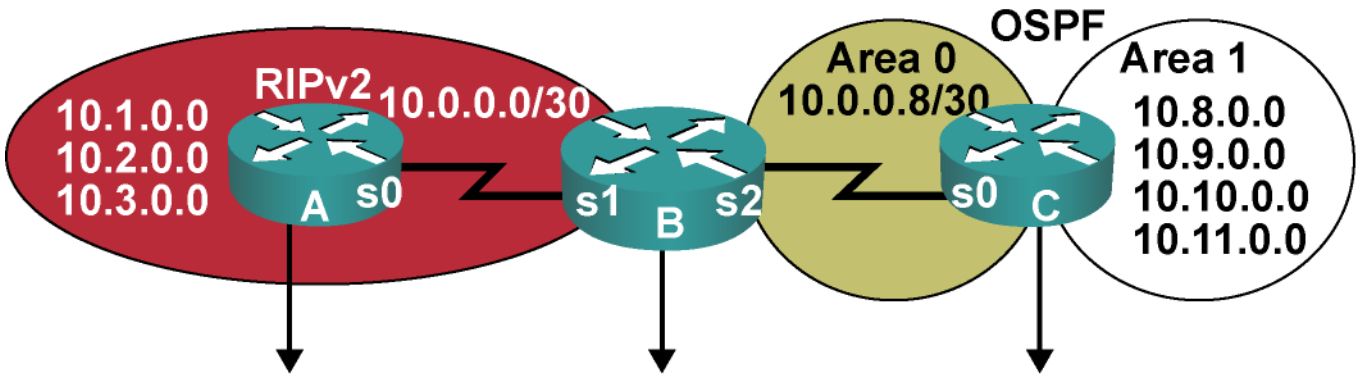
Example: Routing Tables After Route Redistribution



Example: Routing Tables After Summarizing Routes and Redistributions

```
RouterA(config)#interface s0
RouterA(config-if)#ip summary-address rip 10.0.0.0 255.252.0.0
```

```
RouterC(config)#router ospf 1
RouterC(config-router)#area 1 range 10.8.0.0 255.252.0.0
```



A Routing Table

```
C       10.0.0.0
R       10.1.0.0
R       10.2.0.0
R       10.3.0.0
R       10.8.0.0/14
```

B Routing Table

```
C       10.0.0.0
C       10.0.0.8
R       10.0.0.0/14
O IA    10.8.0.0/14
```

C Routing Table

```
C       10.0.0.8
O E2    10.0.0.0/14
O       10.8.0.0
O       10.9.0.0
O       10.10.0.0
O       10.11.0.0
```

0146_316

Summary

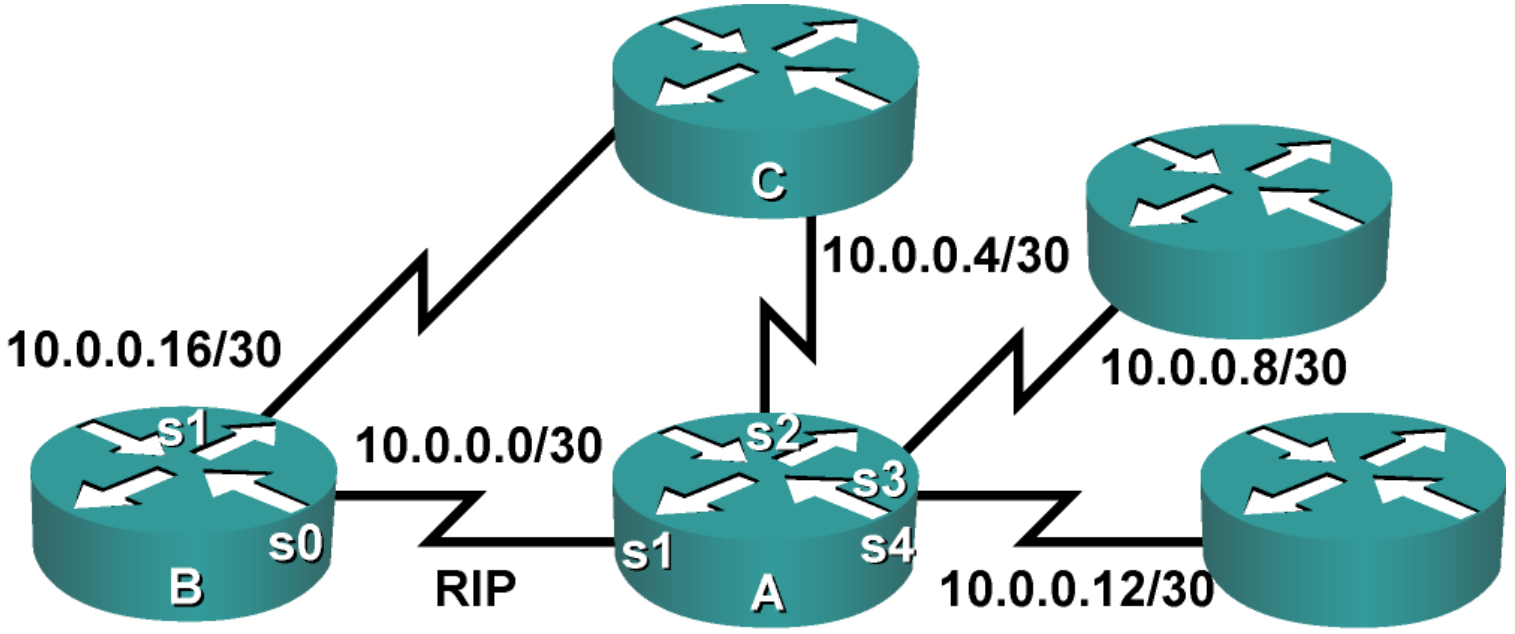
- **Several steps must be followed for accurate IP route redistribution to occur.**
- **All IP routing protocols can be redistributed into RIP.**
- **When IP routing protocols are redistributed into OSPF, additional commands are required.**
- **When IP routing protocols are redistributed into EIGRP, a seed metric is required.**
- **IP routing protocols are usually redistributed into IS-IS as Level 2 routes.**
- **There are several techniques for verifying IP route redistribution.**



Manipulating Routing Updates

Controlling Routing Update Traffic

Using the passive-interface Command



Router B Configuration

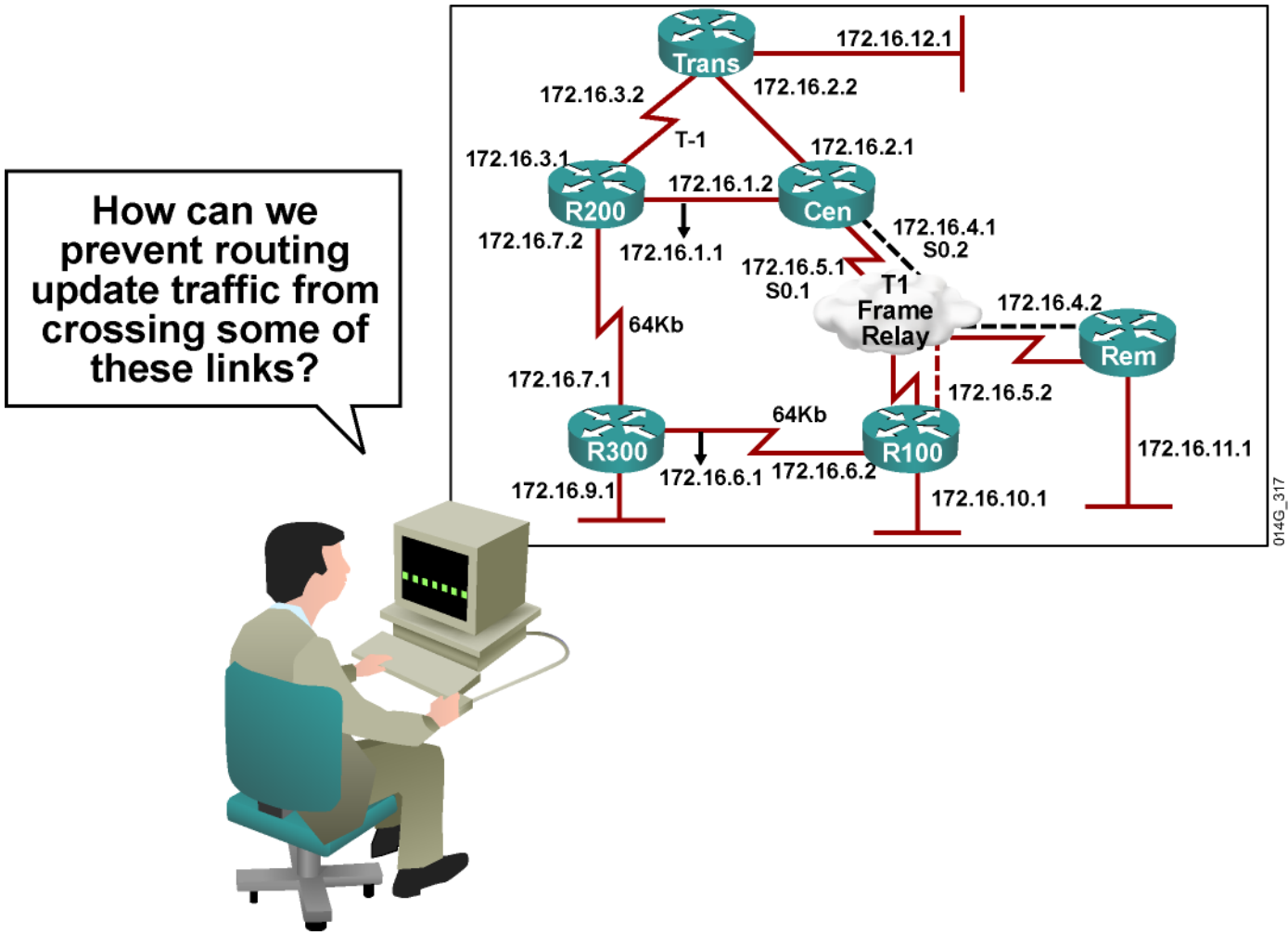
```
router rip
  network 10.0.0.0
  passive-interface s1
```

Router A Configuration

```
router rip
  network 10.0.0.0
  passive-interface default
  no passive-interface s1
```

014G_292

Controlling Routing Update Traffic



Configuring distribute-list

For outbound updates:

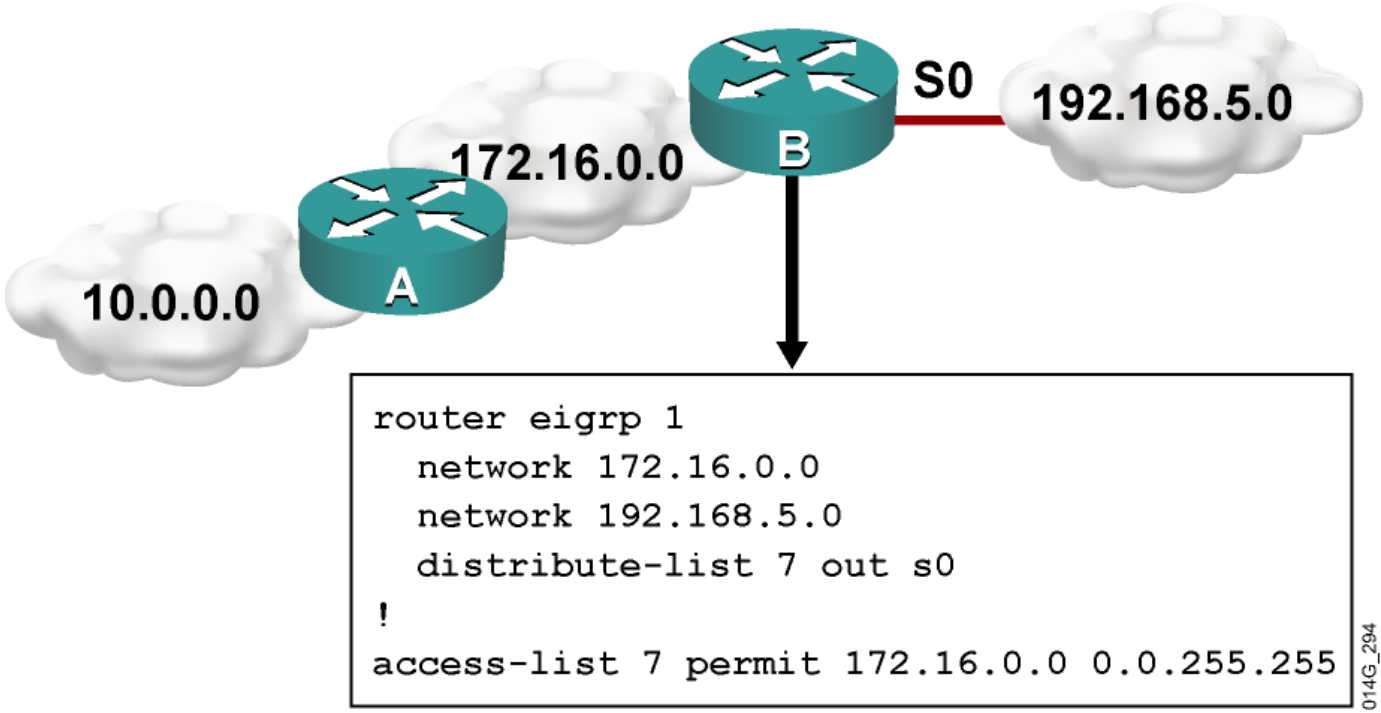
```
Router(config-router)#
distribute-list {access-list-number | name} out
[interface-name  | routing-process [routing-process
parameter]]
```

For inbound updates:

```
Router(config-router)#
distribute-list [access-list-number | name] | [route-map
map-tag] in [interface-type interface-number]]
```

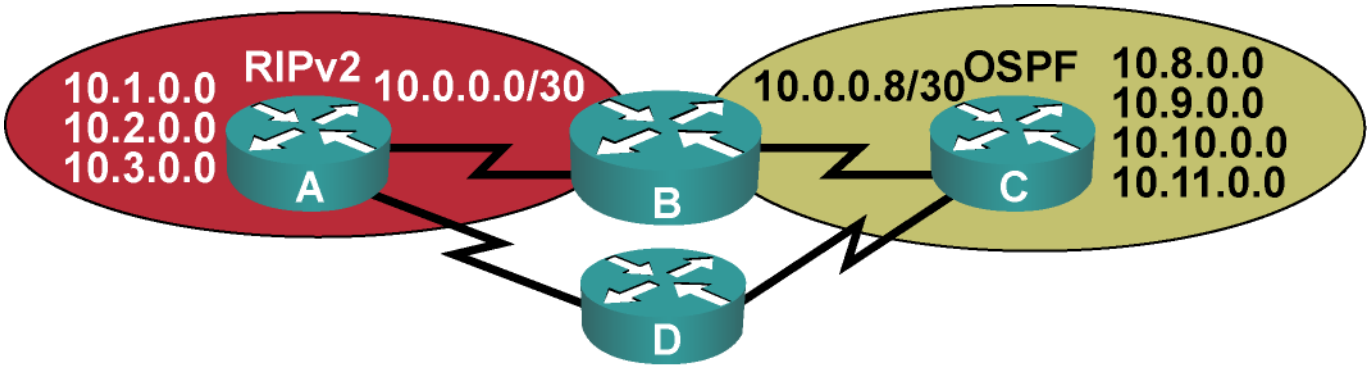
- Use an access list (or route map) to permit or deny routes.
- Can be applied to transmitted, received, or redistributed routing updates.

Filtering Routing Updates with a Distribute List



- **Hides network 10.0.0.0 using interface filtering**

Controlling Redistribution with Distribute Lists



Router B Configuration

```
router ospf 1
  network 10.0.0.8 0.0.0.3 area 0
  redistribute rip subnets
  distribute-list 2 out rip

router rip
  network 10.0.0.0
  version 2
  passive-interface s3
  redistribute ospf 1 metric 5
  distribute-list 3 out ospf 1

access-list 2 permit 10.0.0.0 0.3.255.255

access-list 3 permit 10.8.0.0 0.3.255.255
```

314P_163

Route Maps

Route maps are similar to a scripting language for these reasons:

- **They work like a more sophisticated access list.**
  - **They offer top-down processing.**
  - **Once there is a match, leave the route map.**
- **Lines are sequence-numbered for easier editing.**
  - **Insertion of lines**
  - **Deletion of lines**
- **Route maps are named rather than numbered for easier documentation.**
- **Match criteria and set criteria can be used, similar to the “if, then” logic in a scripting language.**

Route Map Applications

The common uses of route maps are as follows:

- **Redistribution route filtering: a more sophisticated alternative to distribute lists**
- **Policy-based routing: the ability to determine routing policy based on criteria other than the destination network**
- **BGP policy implementation: the primary tool for defining BGP routing policies**

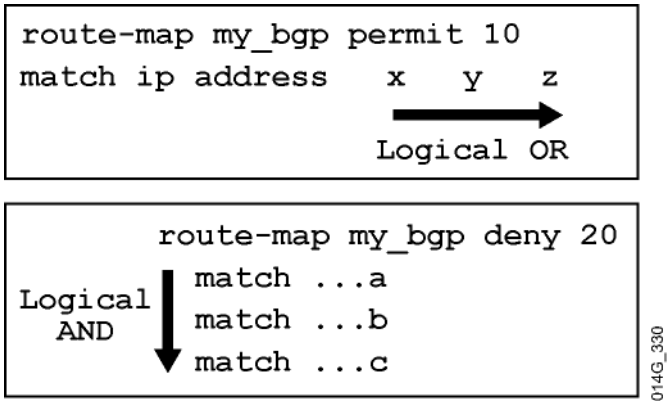
Route Map Operation

- **A list of statements constitutes a route map.**
- **The list is processed top-down like an access list.**
- **The first match found for a route is applied.**
- **The sequence number is used for inserting or deleting specific route map statements.**

```
route-map my_bgp permit 10
       { match statements }
       { match statements }
       { set statements }
       { set statements }
route-map my_bgp deny 20
       ::    ::    ::
       ::    ::    ::
route-map my_bgp permit 30
       ::    ::    ::
       ::    ::    ::
```

Route Map Operation (Cont.)

- The match statement may contain multiple references.
- Multiple match criteria in the same line use a logical OR.
- At least one reference must permit the route for it to be a candidate for redistribution.



- Each vertical match uses a logical AND.
- All match statements must permit the route for it to remain a candidate for redistribution.
- Route map permit or deny determines if the candidate will be redistributed.

route-map Commands

```
router(config)#
route-map map-tag [permit | deny] [sequence-number]
```

- **Defines the route map conditions**

```
router(config-route-map)#
match {conditions}
```

- **Defines the conditions to match**

```
router(config-route-map)#
set {actions}
```

- **Defines the action to be taken on a match**

```
router(config-router)#
redistribute protocol [process id] route-map map-tag
```

- **Allows for detailed control of routes being redistributed into a routing protocol**

The match Command

```
router(config-route-map)#
```

- **The** match **commands specify criteria to be matched.**
- **The associated route map statement permits or denies the matching routes.**

```
Match {options}

  options :
  ip address ip-access-list
  ip route-source ip-access-list
  ip next-hop ip-access-list
  interface type number
  metric metric-value
  route-type [external | internal | level-1 | level-2 |local]
  …
```

The set Command

```
router(config-route-map)#
```

- **The** set **commands modify matching routes.**
- **The command modifies parameters in redistributed routes.**

```
set {options}
      options :
      metric metric-value
      metric-type [type-1 | type-2 | internal | external]
      level [level-1 | level-2 | level-1-2 |stub-area | backbone]
      ip next-hop next-hop-address
```

Route Maps and Redistribution Commands

```
Router(config)# router ospf 10
Router(config-router)# redistribute rip route-map redis-rip
```

- **Routes matching either access list 23 or 29 are redistributed with an OSPF cost of 500, external type 1.**
- **Routes permitted by access list 37 are not redistributed.**
- **All other routes are redistributed with an OSPF cost metric of 5000, external type 2.**

```
Router(config)#
route-map redis-rip permit 10
match ip address 23  29
set metric 500
set metric-type type-1

route-map redis-rip deny 20
match ip address 37

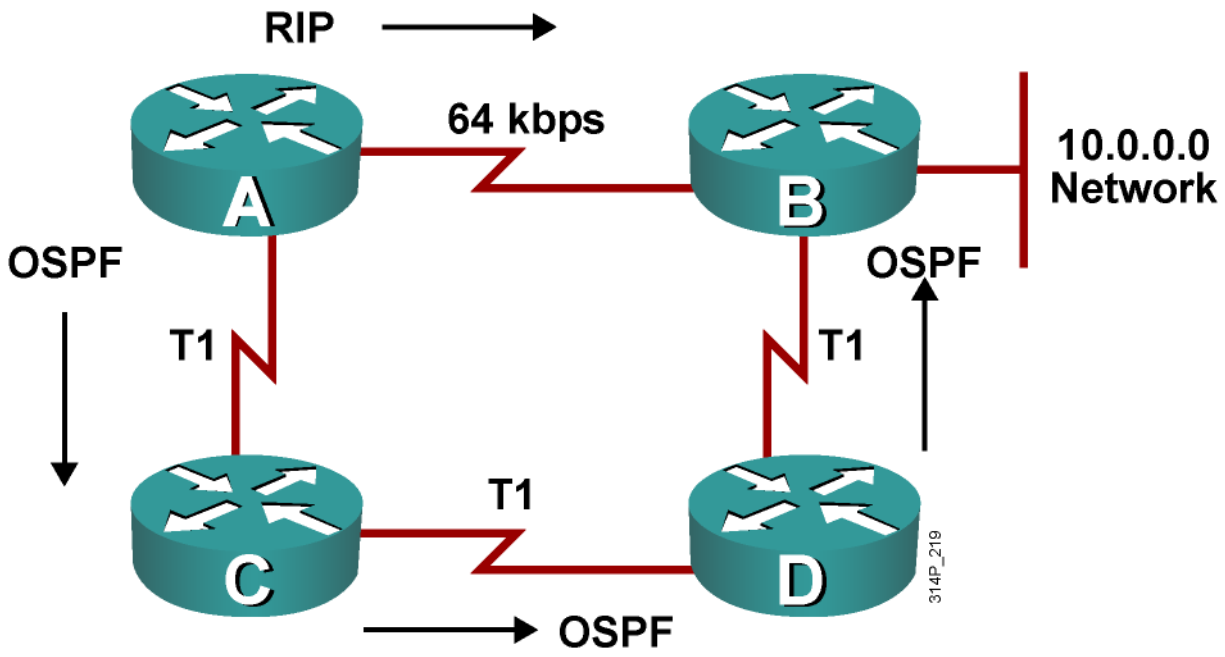
route-map redis-rip permit 30
set metric 5000
set metric-type type-2
```

```
Router(config)#
access-list 23 permit 10.1.0.0 0.0.255.255
access-list 29 permit 172.16.1.0 0.0.0.255
access-list 37 permit 10.0.0.0 0.255.255.255
```

Administrative Distance

Route Source	Default Distance
Connected interface	0
Static route	1
EIGRP summary route	5
External BGP	20
Internal EIGRP	90
IGRP	100
OSPF	110
IS-IS	115
RIPv1, RIPv2	120
External EIGRP	170
Internal BGP	200
Unknown	255

Administrative Distance (Cont.)



Modifying Administrative Distance

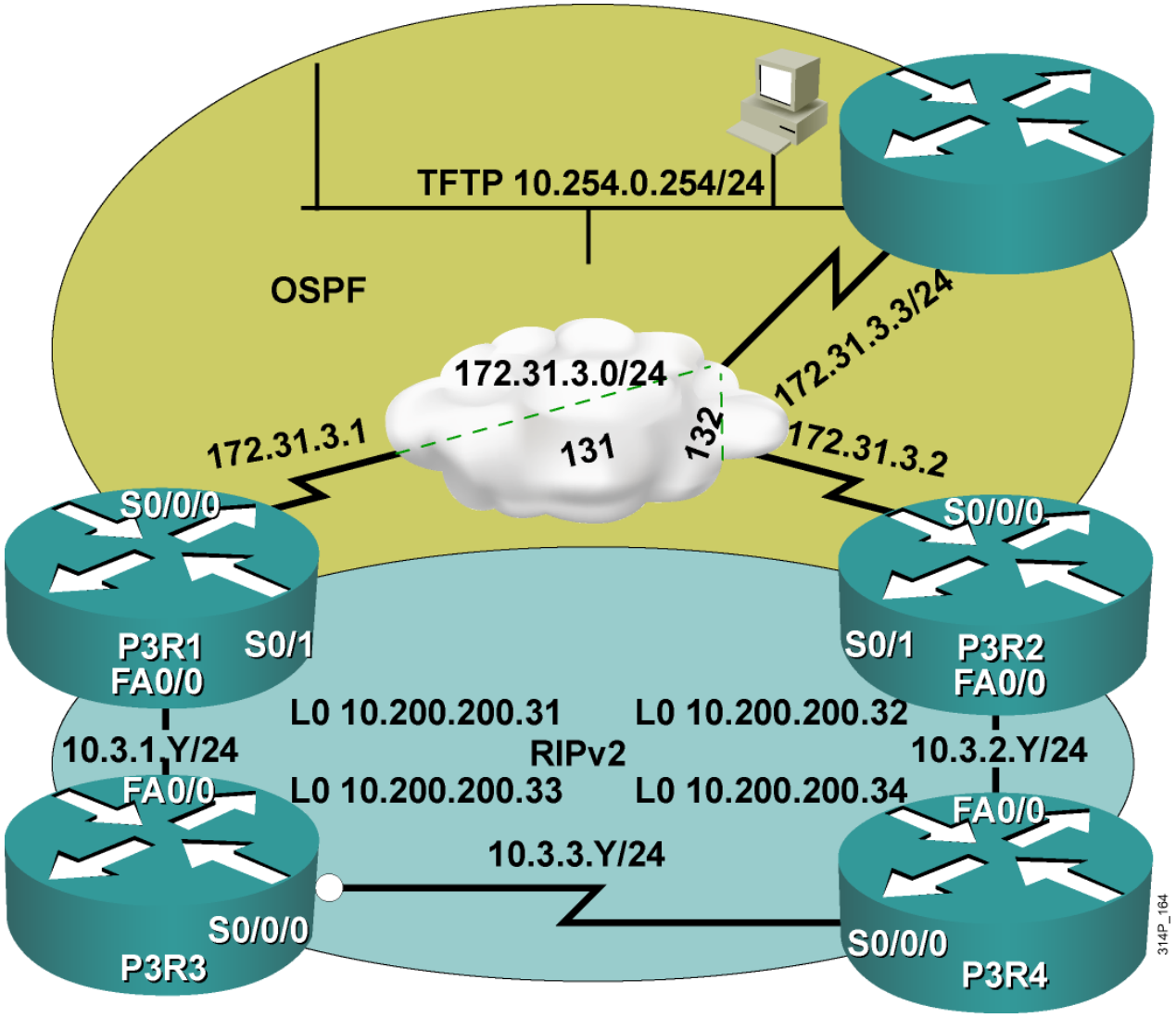
```
Router(config-router)#
 distance administrative distance [address wildcard-mask
[access-list-number | name]]
```

- **Used for all protocols except EIGRP and BGP redistribution**

```
Router(config-router)#
distance eigrp internal-distance external-distance
```

- **Used for EIGRP**

Example: Redistribution Using Administrative Distance



Example: Redistribution Using Administrative Distance (Cont.)

Router P3R1

```
router ospf 1
 redistribute rip metric 10000 metric-type 1 subnets
 network 172.31.0.0 0.0.255.255 area 0
!
router rip
 version 2
 redistribute ospf 1 metric 5
 network 10.0.0.0
 no auto-summary
```

Router P3R2

```
router ospf 1
 redistribute rip metric 10000 metric-type 1 subnets
 network 172.31.3.2 0.0.0.0 area 0
!
router rip
 version 2
 redistribute ospf 1 metric 5
 network 10.0.0.0
 no auto-summary
```

Example: Redistribution Using Administrative Distance (Cont.)

With OSPF and RIP running




```
P3R2#show ip route
<Output Omitted>

Gateway of last resort is not set

     172.31.0.0/24 is subnetted, 7 subnets
O       172.31.55.0 [110/2343] via 172.31.3.3, 00:09:46, Serial0/0/0
C       172.31.3.0 is directly connected, Serial0/0/0
O       172.31.2.0 [110/1562] via 172.31.3.3, 00:09:46, Serial0/0/0
10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O E1    10.3.1.0/24 [110/10781] via 172.31.3.1, 00:09:47, Serial0/0/0
O E1    10.3.3.0/24 [110/10781] via 172.31.3.1, 00:04:51, Serial0/0/0
C       10.3.2.0/24 is directly connected, fastethernet0/0
O E1    10.200.200.31/32 [110/10781] via 172.31.3.1, 00:09:48, Serial0/0/0
O E1    10.200.200.34/32 [110/10781] via 172.31.3.1, 00:04:52, Serial0/0/0
C       10.200.200.32/32 is directly connected, Loopback0
O E1    10.200.200.33/32 [110/10781] via 172.31.3.1, 00:04:52, Serial0/0/0
O E2    10.254.0.0/24 [110/50] via 172.31.3.3, 00:09:48, Serial0/0/0
```

314P_165

- **P3R2 includes suboptimal paths and loops.**

Example: Redistribution Using Administrative Distance (Cont.)

```
hostname P3R1
!
router ospf 1
 redistribute rip metric 10000 metric-type 1
subnets
 network 172.31.0.0 0.0.255.255 area 0
 distance 125 0.0.0.0 255.255.255.255 64
!
router rip
 version 2
 redistribute ospf 1 metric 5
 network 10.0.0.0
 no auto-summary
!
access-list 64 permit 10.3.1.0 0.0.0.255
access-list 64 permit 10.3.3.0 0.0.0.255
access-list 64 permit 10.3.2.0 0.0.0.255
access-list 64 permit 10.200.200.31
access-list 64 permit 10.200.200.34
access-list 64 permit 10.200.200.32
access-list 64 permit 10.200.200.33
```

```
hostname P3R2
!
router ospf 1
redistribute rip metric 10000 metric-type 1
subnets
 network 172.31.3.2 0.0.0.0 area 0
 distance 125 0.0.0.0 255.255.255.255 64
!
router rip
 version 2
 redistribute ospf 1 metric 5
 network 10.0.0.0
 no auto-summary
!
access-list 64 permit 10.3.1.0 0.0.0.255
access-list 64 permit 10.3.3.0 0.0.0.255
access-list 64 permit 10.3.2.0 0.0.0.255
access-list 64 permit 10.200.200.31
access-list 64 permit 10.200.200.34
access-list 64 permit 10.200.200.32
access-list 64 permit 10.200.200.33
```

Example: Redistribution Using Administrative Distance (Cont.)

With OSPF changing administrative distance




```
Gateway of last resort is not set

     172.31.0.0/16 is variably subnetted, 8 subnets, 2 masks
O       172.31.55.4/32 [110/781] via 172.31.33.4, 00:00:01, Serial0/0/0
C       172.31.33.0/24 is directly connected, Serial0/0/0
O       172.31.33.1/32 [110/1562] via 172.31.33.4, 00:00:01, Serial0/0/0
O       172.31.33.4/32 [110/781] via 172.31.33.4, 00:00:01, Serial0/0/0
O       172.31.44.4/32 [110/781] via 172.31.33.4, 00:00:01, Serial0/0/0
O       172.31.22.4/32 [110/781] via 172.31.33.4, 00:00:01, Serial0/0/0
O       172.31.11.4/32 [110/781] via 172.31.33.4, 00:00:03, Serial0/0/0
O       172.31.66.4/32 [110/781] via 172.31.33.4, 00:00:03, Serial0/0/0
     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
R       10.3.1.0/24 [120/2] via 10.3.2.4, 00:00:03, FastEthernet0/0
R       10.3.3.0/24 [120/1] via 10.3.2.4, 00:00:03, FastEthernet0/0
C       10.3.2.0/24 is directly connected, FastEthernet0/0
R       10.200.200.31/32 [120/3] via 10.3.2.4, 00:00:04, FastEthernet0/0
R       10.200.200.34/32 [120/1] via 10.3.2.4, 00:00:04, FastEthernet0/0
C       10.200.200.32/32 is directly connected, Loopback0
R       10.200.200.33/32 [120/2] via 10.3.2.4, 00:00:04, FastEthernet0/0
O E2    10.254.0.0/24 [110/50] via 172.31.33.4, 00:00:04, Serial0/0/0
```

314P_166

- **Router P3R2 prefers RIP routes.**

Summary

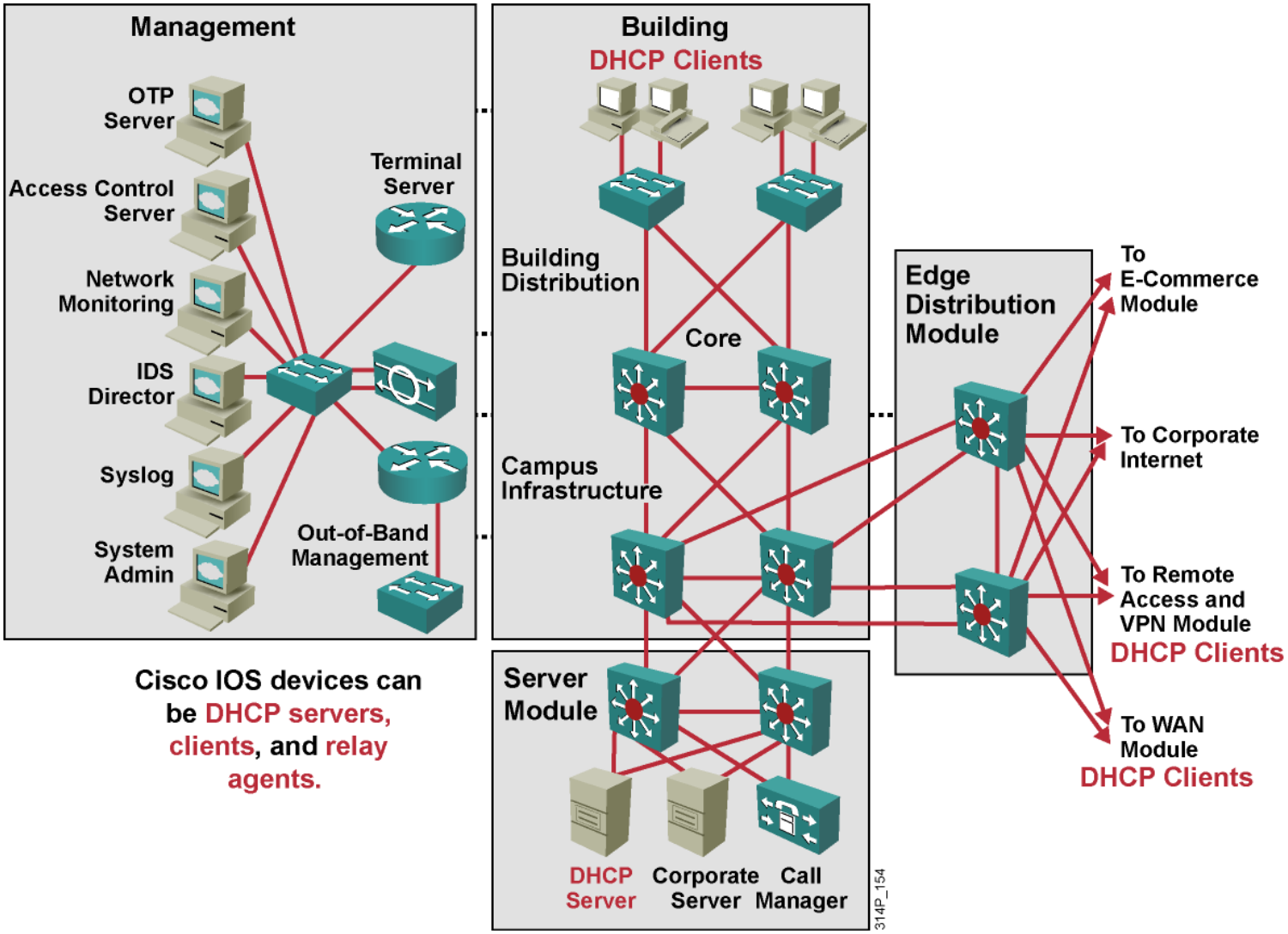
- **The** passive-interface **command allows control of routing updates.**
- **A distribute list uses an ACL to control routing updates.**
- **A distribute list may be applied to an interface or to redistribute routes.**
- **A route map is a complex tool used for manipulating and filtering routes and uses match–set or if–then logic.**
- **A route-map can be used to streamline the route redistribution process.**
- **Administrative distance is a value used by routers to evaluate the route received from more than one routing protocol.**
- **Each IP routing protocol is assigned a value by Cisco, which can be changed with Cisco IOS software commands.**
- **During route redistribution, administrative distance must be manipulated at times to maintain routing accuracy.**



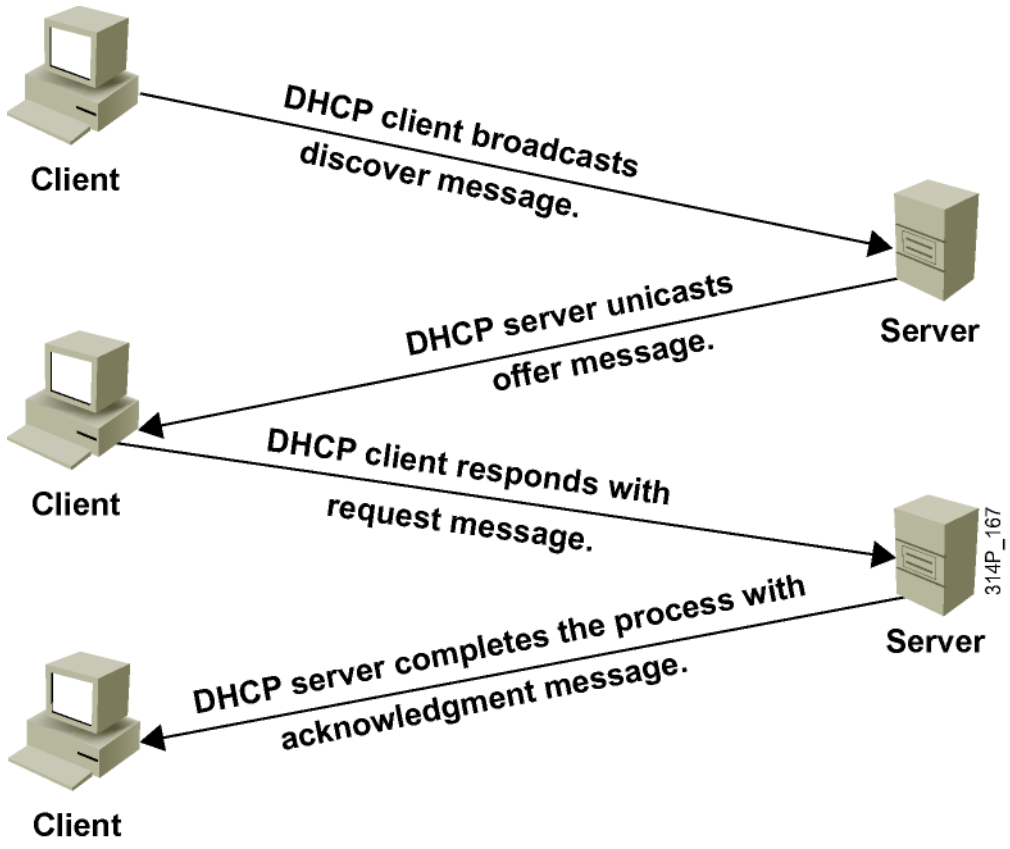
Manipulating Routing Updates

Implementing Advanced Cisco IOS Features: Configuring DHCP

DHCP in an Enterprise Network



Dynamic Host Configuration Protocol



Configuring a DHCP Server

```
Router(config)#ip dhcp pool [pool name]
```

- **Enables a DHCP pool for use by hosts**

```
Router(config-dhcp)#import all
```

- **Imports DNS and WINS information from IPCP**

```
Router(config-dhcp)#network [network address][subnet mask]
```

- **Specifies the network and subnet mask of the pool**

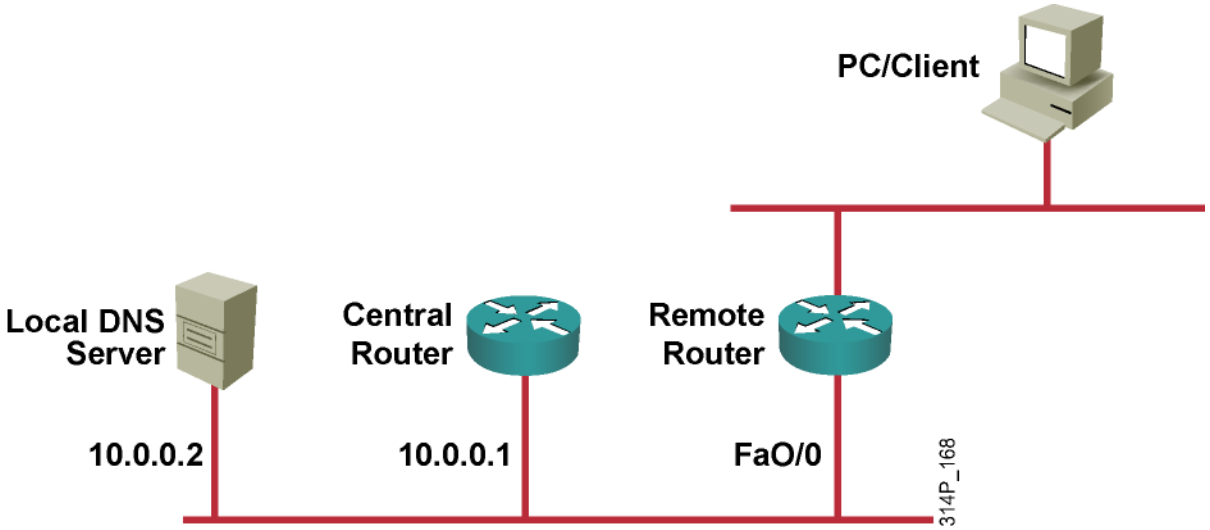
```
Router(config-dhcp)#default-router [host address]
```

- **Specifies the default router for the pool to use**

DHCP Server Configuration Example

```
ipdhcp database ftp://user:passwords@172.16.4.253/router-dhcp write-delay 120
ip dhcp excluded-address 172.16.1.100 172.16.1.103
ip dhcp excluded-address 172.16.2.100 172.16.2.103
ip dhcp pool 0
  network 172.16.0.0/16
  domain-name global.com
  dns-server 172.16.1.102 172.16.2.102
  netbios-name-server 172.16.2.103 172.16.2.103
  default-router 172.16.1.100
```

Importing and Autoconfiguration



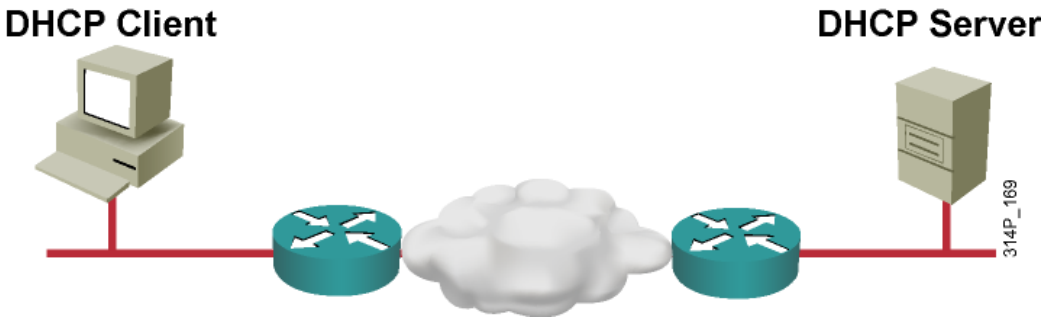
DHCP Client

```
Router (config-if)#
```

```
ip address dhcp
```

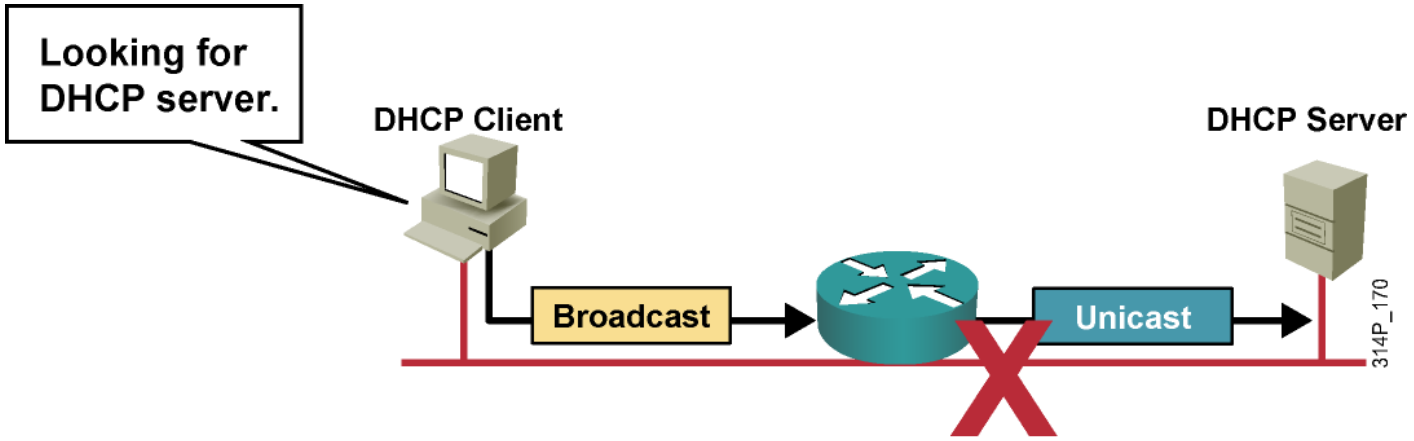
Enables a Cisco IOS device to obtain an IP address dynamically from a DHCP server

Helper Addressing Overview



- **Routers do not forward broadcasts, by default.**
- **Helper address provides selective connectivity.**

Why Use a Helper Address?



- **Sometimes clients do not know the server address.**
- **Helpers change broadcast to unicast to reach server.**

IP Helper Address Commands

Router(config-if)#

ip helper-address *address*

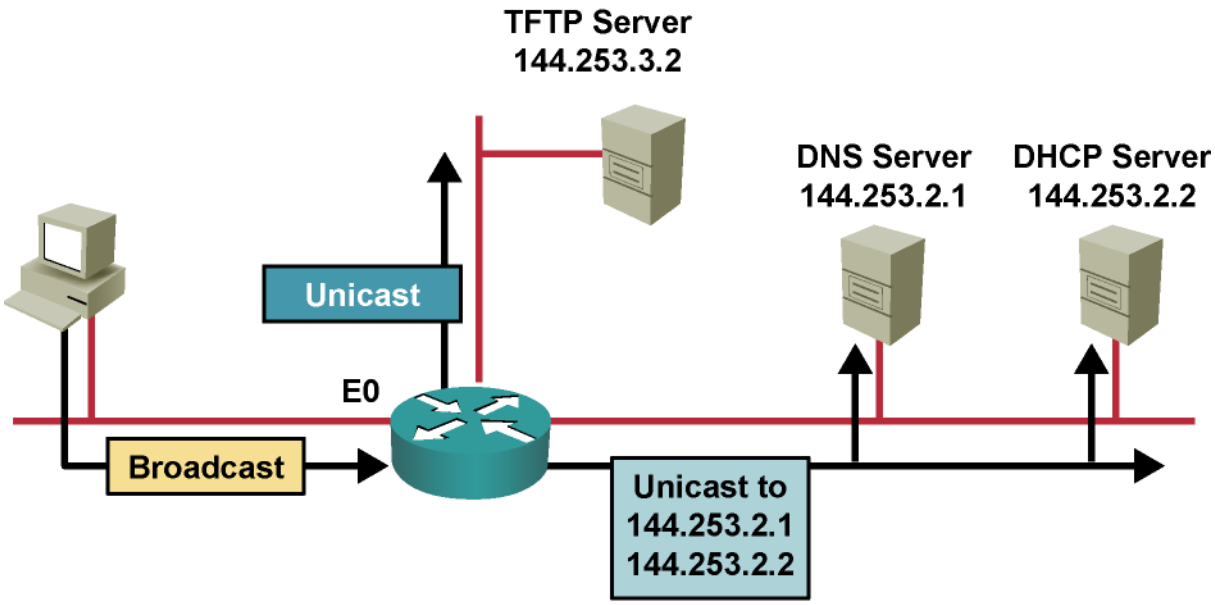
- **Enables forwarding and specifies destination address for main UDP broadcast packets**
- **Changes destination address from broadcast to unicast or directed broadcast address**

Router(config)#

ip forward-protocol { udp [*port*] }

- **Specifies which protocols will be forwarded**

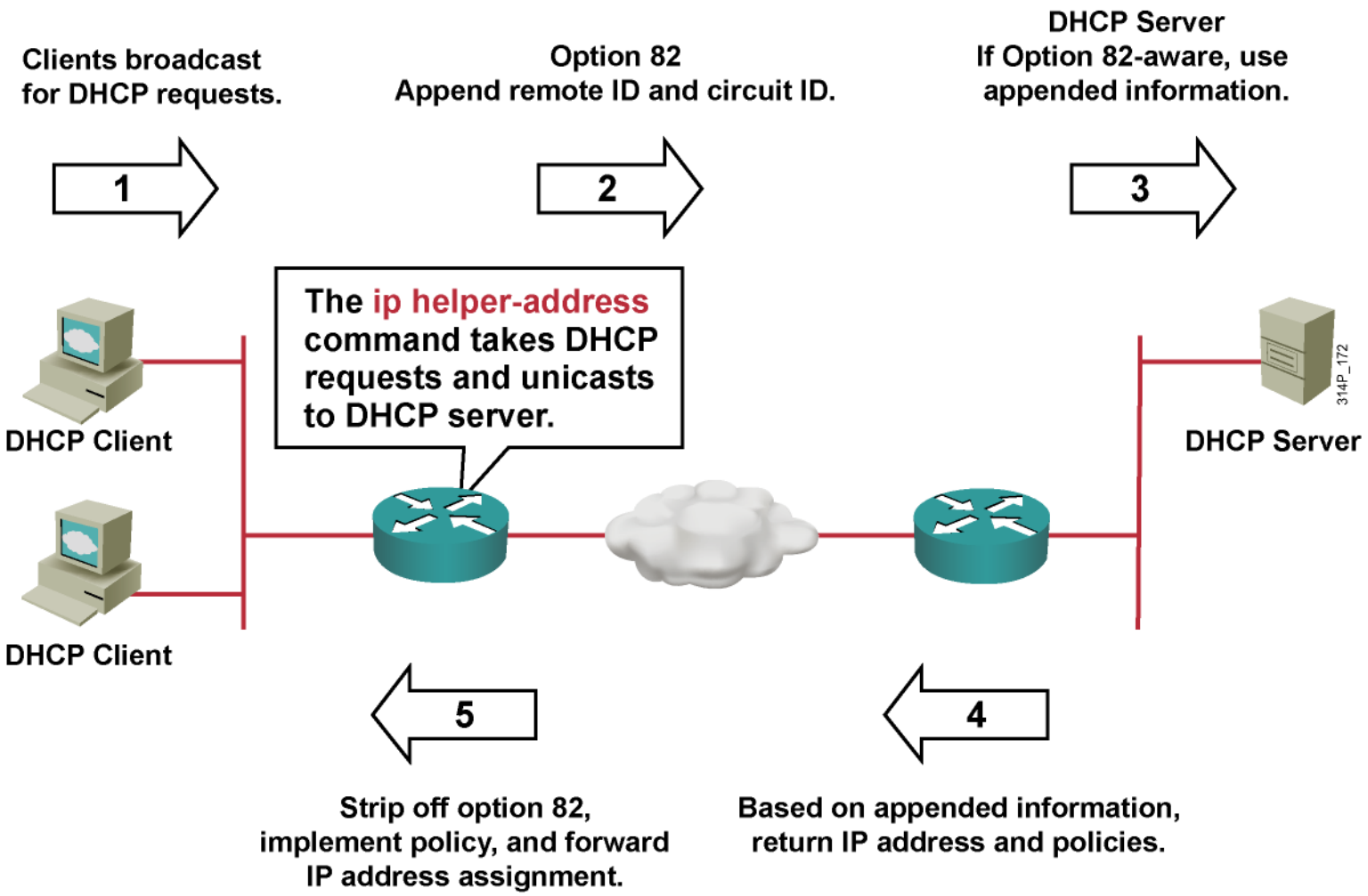
Multiple Servers: Remote Networks




```
interface ethernet 0
ip address 144.253.1.100  255.255.0
ip helper-address 144.253.2.1
ip helper-address 144.253.2.2
ip helper-address 144.253.3.2
```

314P_171

Relay Agent Option Support



DHCP Verification Commands

```
router#
show ip dhcp database
```

- Displays recent activity on the DHCP database

```
router#
show ip dhcp server statistics
```

- Shows count information about statistics and messages sent and received

```
router#
show ip route dhcp
```

- Displays routes added to the routing table by DHCP

```
router#
debug ip dhcp server {events | packets | linkage}
```

- Enables debugging on the DHCP server

Summary

- **DHCP functions may be configured with Cisco IOS software.**
- **DHCP server can be configured.**
- **DHCP options can be configured.**
- **DHCP client can be configured.**
- **The IP helper address activates the DHCP relay agent in the Cisco IOS device.**
- **DHCP relay services are supported.**

Module Summary

- **Using multiple IP routing protocols can be a result of migrating to a more advanced routing protocol, a multivendor environment, political boundaries, or device mismatch.**
- **Route redistribution is possible between any two IP routing protocols.**
- **It is necessary to use filters and modify administrative distance during redistribution.**
- **A Cisco IOS device can be configured as a DHCP server, relay agent, or client.**


CISCO SYSTEMS